

Bruce Jennings

Electronic & Computer Engineering Student

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Professional Profile

Electronic and Computer Engineering student entering the final year of a BEng at the University of Nottingham, with hands-on experience in embedded systems, FPGA communication, sensor interfacing and C++ software development.

Interested in developing intelligent electronic systems that combine hardware, software, sensing and signal processing, with interests in embedded systems, robotics, edge AI and medical technology.

Seeking graduate, internship and technical project opportunities in embedded systems, software and applied electronics.

Education

University of Nottingham – BEng (Hons) Electronic and Computer Engineering | *Sept 2024 – Present*

- On track for a First-Class Honours degree, with relevant modules in embedded systems, signal processing, control, power electronics, software development, and mathematical modelling.

University of Nottingham – Engineering and Physical Sciences Foundation Year | *Sept 2023 – June 2024* | 2:1

Technical Skills

- Programming & Firmware:** C, C++, Python, MATLAB
- Embedded & Hardware:** STM32, Arduino, UART, PWM, sensor interfacing
- Tools & Software:** STM32CubeIDE, Quartus Prime, LTspice, KiCad, Git, GitHub, CMake, Qt, Node-RED
- Engineering Practices:** Oscilloscope debugging, circuit testing, soldering, technical documentation, system validation

Engineering Project Experience

Line-Following Robot | Individual Project

- Built an autonomous Arduino robot using infrared sensors for real-time path detection.
- Implemented PID motor control and a Node-RED dashboard, refining performance through sensor calibration.

Pulse Oximeter Device | Group Project

- Contributed to a real-time pulse oximeter prototype using STM32 signal acquisition and FPGA-based digital logic.
- Designed UART communication in Quartus Prime and used oscilloscopes to validate timing and hardware-software integration.

Virtual Reality CAD Visualisation System | Group Project

- Developed a C++/Qt CAD viewer using VTK and OpenVR to visualise STL assemblies in desktop and VR environments.
- Implemented recursive STL loading, tree-based model controls, live GUI-to-VR updates, safe VR threading, Doxygen documentation, and installer packaging.

Experience

Augimono, University of Nottingham | Consulting Micro-Placement | June – July 2026

- Produced a consultancy-grade client report on adapting gamified learning tools for workforce readiness, with recommendations now informing Augimono's product language, measurement framework and next product release.

TK Maxx | Retail Associate | Nov 2022 – Sept 2025

- Balanced 16+ hours of weekly work alongside university; mentored 8+ new staff members on POS systems and store procedures.

Elite Personnel LTD | Administration Assistant | June 2019